

Customer Feedback Theme Miner and Action Recommender

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1. Project Overview

Project Title

Customer Feedback Theme Miner and Action Recommender

Objective

The objective of this project is to analyze customer survey comments, complaints, support tickets, and feedback messages to identify recurring themes and generate actionable recommendations for product and service improvement.

The system leverages Agentic AI workflows, OpenAI-powered semantic analysis, and ASP.NET Core Web APIs to automate feedback intelligence generation.

2. Problem Statement

Organizations receive large volumes of unstructured customer feedback from multiple channels such as CSAT surveys, NPS responses, support tickets, and escalation reports. Manual analysis of this feedback is time-consuming, inconsistent, and difficult to scale.

Traditional approaches often fail to:

- Detect recurring issue patterns efficiently
- Prioritize high-impact customer concerns
- Generate actionable improvement recommendations
- Maintain consistent theme categorization
- Provide explainable insights to business teams

This project addresses these challenges by implementing an AI-driven feedback mining and recommendation platform.

3. Project Scope

In Scope

- Customer feedback ingestion
- Feedback preprocessing and cleaning
- Embedding-based issue clustering
- AI-powered theme labeling
- Sentiment and urgency extraction
- Action recommendation generation
- SQL Server-based data storage
- REST API development using ASP.NET Core
- Analytics and reporting support
- Weekly feedback digest generation
- Cluster export functionality

4. Users / Actors

User Type	Responsibilities
Customer Success Teams	Analyze customer pain points and recurring complaints
Product Managers	Identify feature improvement opportunities
Quality Teams	Track quality-related issues and recommendations
Operations Teams	Monitor escalations and service impact

5. Inputs and Data Source

The system processes multiple categories of customer feedback data.

Data Inputs

- CSAT comments
- NPS feedback responses
- Support feedback and complaint logs
- Product feature feedback
- Escalation categories
- Customer issue descriptions

Data Characteristics

- Semi-structured text data
- Multi-source feedback inputs
- Repetitive and duplicate issue patterns
- Sentiment-rich customer narratives

6. System Architecture

Technology Stack

Technology	Purpose
ASP.NET Core Web API	Backend API development
C#	Business logic and orchestration
OpenAI API	NLP analysis and recommendation generation
SQL Server	Centralized data storage
SSMS	Database management
VS Code	Development environment (Angular)
GitHub	Version control and collaboration
Swagger	API testing and documentation

7. Core AI Components

Embedding-Based Clustering

Semantic embeddings are used to group similar customer issues together. This helps identify duplicate complaints and recurring themes across large feedback datasets.

GPT Theme Labeling

The OpenAI GPT model generates meaningful and consistent labels for clustered customer issues.

Function Calling for Action Classification

The system classifies recommended actions based on detected customer pain points and business impact.

Sentiment and Urgency Extraction

AI models evaluate customer sentiment and urgency levels to prioritize critical issues.

8. Workflow Architecture

End-to-End Workflow

Ingest Feedback → Cluster Similar Issues → Label Themes → Measure Impact → Recommend Action Items → Generate Analytics

Workflow Description

Step 1 — Feedback Ingestion

Customer comments and survey feedback are uploaded through the API layer and stored in SQL Server.

Step 2 — Semantic Clustering

Embedding-based clustering groups similar customer issues together.

Step 3 — Theme Detection

OpenAI APIs generate semantic theme labels for identified clusters.

Step 4 — Impact Analysis

The system evaluates issue frequency, urgency, and business impact.

Step 5 — Recommendation Generation

Action recommendations are generated using AI-assisted classification of workflows.

Step 6 — Analytics & Reporting

The system prepares summarized insights, dashboards, and exportable reports.

9. Database Design

Core Database Tables

Table Name	Purpose
Feedback Items	Stores customer feedback entries
Theme Clusters	Stores clustered feedback groups
Themes	Stores generated theme labels
Processing Runs	Tracks processing executions
Action Recommendations	Stores generated recommendations
Evaluation Runs	Stores evaluation metrics
Theme Evaluations	Stores theme validation results
Scheduled Digest Runs	Stores weekly digest execution logs

10. API Specifications

Implemented APIs

API Endpoint	Method	Purpose
/upload-feedback	POST	Upload customer feedback
/analyze	POST	Trigger AI analysis workflow
/themes	GET	Retrieve generated themes
/recommendations	GET	Retrieve recommended actions

API Characteristics

- RESTful architecture
- JSON-based communication
- Modular service integration
- Scalable processing workflow

11. Nonfunctional Requirements

PII Redaction

Sensitive customer information is masked before AI processing to maintain privacy and compliance.

Multi- Linguistic Support

The miner supports feedback from multiple languages.

Explainable Clustering Output

Cluster outputs are designed to remain interpretable and understandable for business users.

Stable Theme Naming Across Runs

The system maintains consistency in generated theme labels across repeated executions.

12. Evaluation Metrics Achieved

Evaluation Metric	Target
Theme Relevance	$\geq 4/5$
Duplicate Issue Clustering Precision	≥ 0.8
Action Recommendation Usefulness	$\geq 4/5$

13. Deliverables

Final Deliverables

- Theme dashboard
- Weekly feedback digest
- CSV report generator
- Evaluation notebook
- REST APIs

14. Setup Instructions

Prerequisites

- .NET SDK installed
- SQL Server installed
- SSMS installed
- Visual Studio Code installed
- OpenAI API key configured
- GitHub repository access

BACKEND

Step 1 — Configure Database

Update connection strings with your server's name.

Step 3 — Configure OpenAI API

Add the OpenAI API key into the .env file.

Step 4 — Build and Run Application

Run the ASP.NET Core Web API project using the .NET CLI.

Step 5 — Verify APIs

Use Swagger to test implemented API endpoints.

FRONTEND

UI setup and run

1. Install Node.js 20+ and Angular CLI if not already installed:

```
npm install -g @angular/cli
```

2. Go to the Angular dashboard folder:

```
cd angular-dashboard
```

3. Install UI dependencies:

```
npm install
```

4. Start the Angular development server:

```
npm start
```

5. Open the app in browser:

<http://localhost:4200>

6. Steps to run the application

1. Add Feedback (Feedback Page)

- Enter customer feedback items one by one
- System auto-refreshes the list

2. Run Analysis (Themes Page)

- Click "Run Analysis" button

- AI clusters similar feedback & extracts themes
- Takes 4-5 minutes

3. Evaluate the metrics

- Click on Evaluate button in the reports section.

4. View Themes (Themes Page)

- See extracted themes with:
 - Descriptions
 - Feedback count
 - Sentiment/Urgency scores
 - Impact scores
 - Top recommendations

5. View Dashboard (Dashboard Page)

- Key metrics cards
- Sentiment breakdown
- Top themes by impact
- Action recommendations
- Evaluation targets
- Recent feedback list
- Ask assistant

6. Generate Reports (Reports Page)

- Download Weekly Digest CSV
- View/evaluate processing runs
- Export evaluation notebooks (HTML/JSON)
- Export feedback clusters

1. Submit Feedback

1. Submit Feedback

FeedbackMiner

Main

- Dashboard
- Feedback

Analysis

- Themes
- Reports

FeedbackMiner

INGESTION

Feedback

Add Feedback

Feedback typeManual

Feedback text?app is useful

+ Save Feedback

Feedback Items

Newest firstAll sentiment

21 shown

manual

Checkout and payment processing feel much faster after the recent application update.

English

Unscreened

survey

O login falha após redefinir minha senha. Meu IP é REDACTED_IP

Portuguese

Unscreened

survey

My name is REDACTED_NAME. Payment keeps failing during checkout even though my card works on other websites. Contact me at REDACTED_EMAIL

English

Unscreened

review

The dashboard interface looks different after the recent update but functionality remains the same.

English

Unscreened

support

No puedo iniciar sesión después de cambiar la contraseña. Mi IP es REDACTED_IP

Spanish

Unscreened

review

Dashboard pages load extremely slowly and navigation feels laggy.

English

Unscreened

manual

Support response times are extremely slow and follow-ups are delayed.

English

Unscreened

2. Open Themes Page

 FeedbackMiner

Main

 Dashboard

 Feedback

Analysis

 Themes

 Reports

FeedbackMiner

ANALYSIS

Themes

☐ New only

☒ All feedback

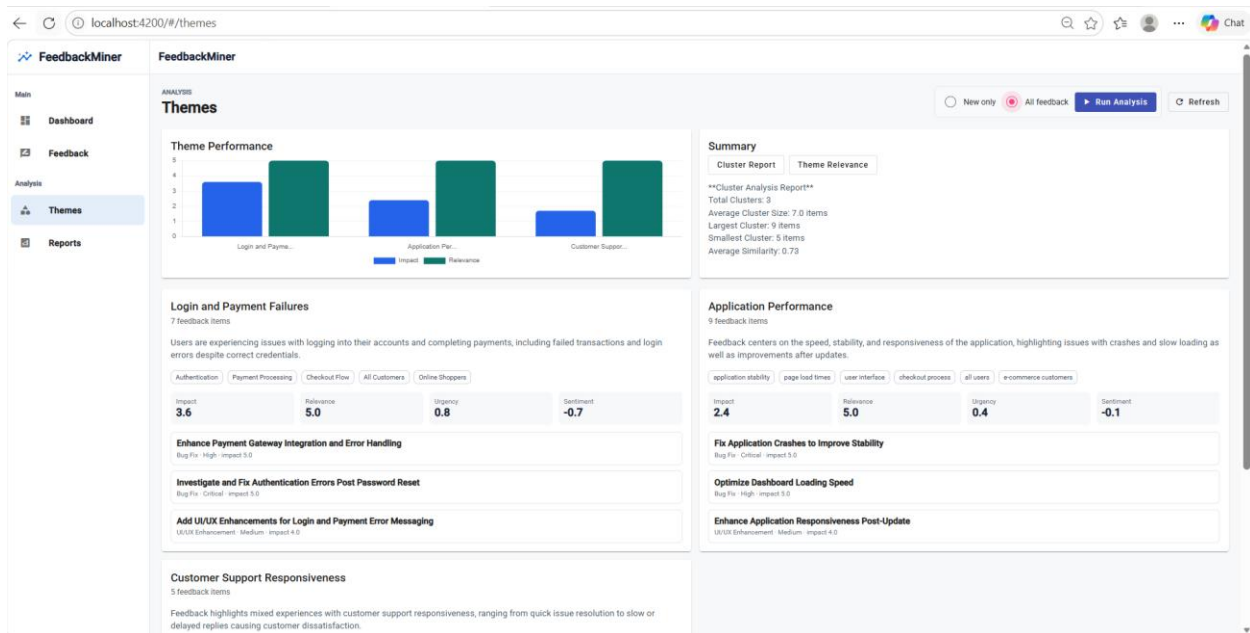
[▶ Run Analysis](#)

[↻ Refresh](#)

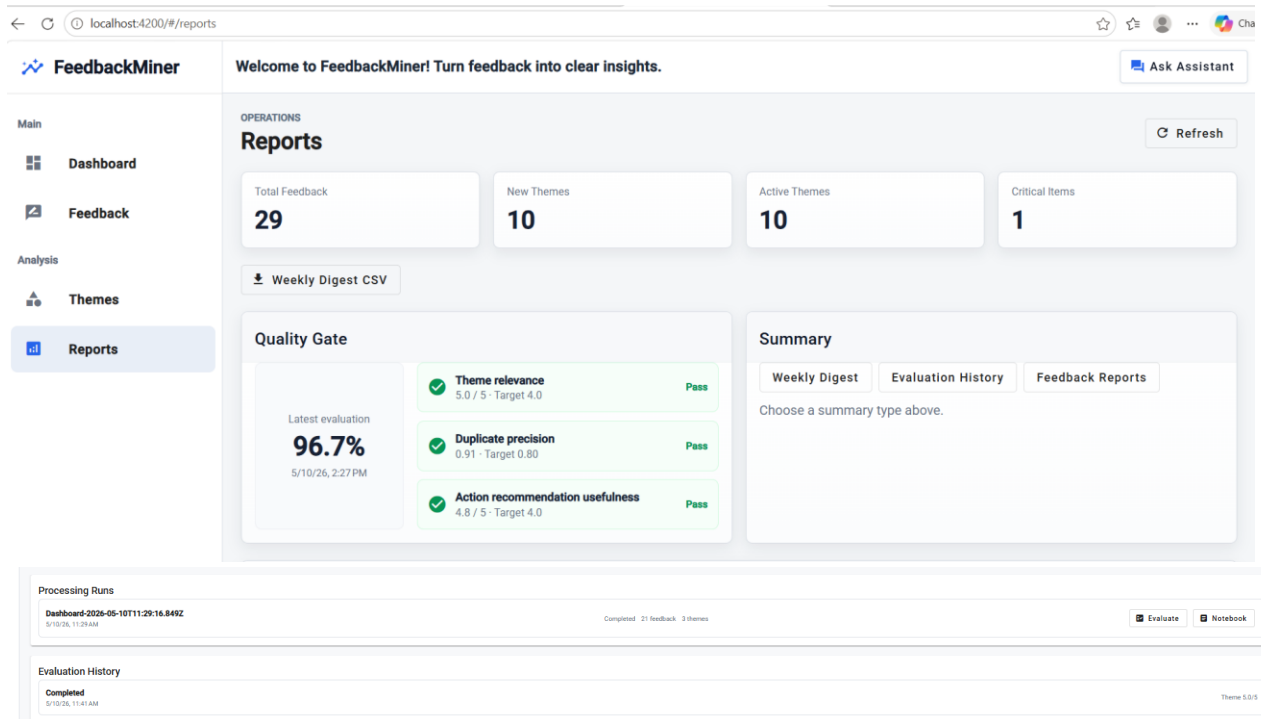
No themes yet

Add feedback, then run analysis to cluster feedback into themes.

3. Click on Run Analysis and wait for themes, clusters creation and action recommendations.



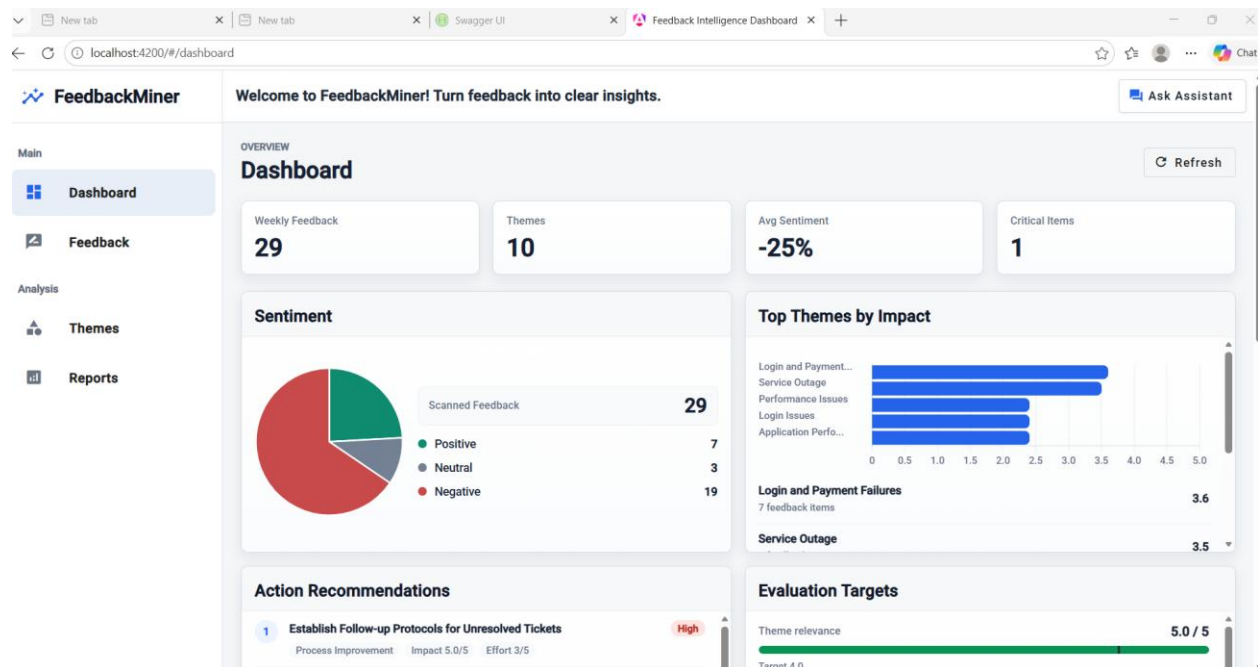
4. Go to the Reports page and click on the Evaluation button. It generates the metrics for theme relevance, cluster precision and action recommendations usefulness.



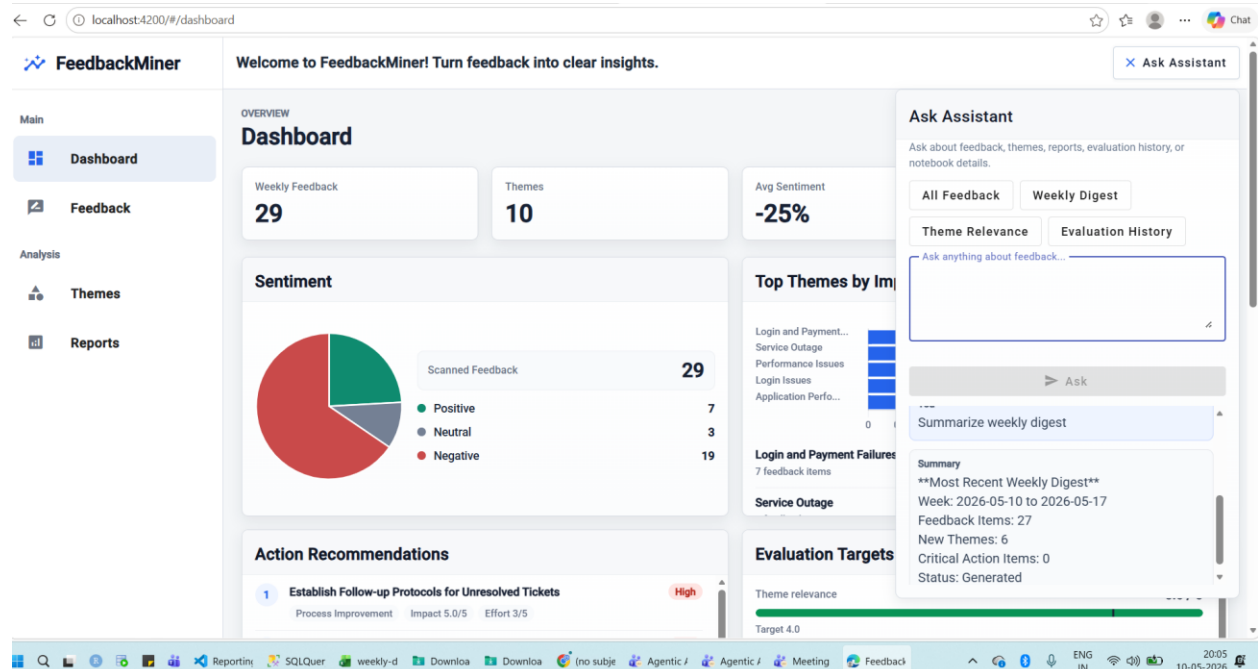
Weekly Digest CSV Report: [weekly-digest-20260510.xlsx](#)

Evaluation Notebook: [evaluation-report-20260510.html](#)

5. Go to the dashboard and check all the populated metrics



6. We also have a chatbot assistant on the top right corner. Click on that to generate metrics summaries.



✕ Ask Assistant

Ask Assistant

Ask about feedback, themes, reports, evaluation history, or notebook details.

All Feedback

Weekly Digest

Theme Relevance

Evaluation History

Ask anything about feedback...

➤ Ask

Summarize weekly digest

Summary

****Most Recent Weekly Digest****
Week: 2026-05-10 to 2026-05-17
Feedback Items: 27
New Themes: 6
Critical Action Items: 0
Status: Generated

16. Conclusion

The Customer Feedback Theme Miner and Action Recommender provide an intelligent and scalable platform for transforming unstructured customer feedback into actionable business intelligence.

By integrating Agentic AI workflows, OpenAI semantic analysis, ASP.NET Core APIs, and SQL Server data management, the system enables organizations to improve customer experience through automated insight generation and recommendation-driven decision-making.